

COMS 4720/5720 Recitation

INTRODUCTION (WEEK-1)

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Grading policy

	Problem sets	Midterm 1	Midterm 2	Final exam	Essay
Undergrad	20 %	22%	25%	33 %	
Grad	20 %	22%	25%	25%	8 %

Late penalty:

1 day late → -10%

2 days late → -20%

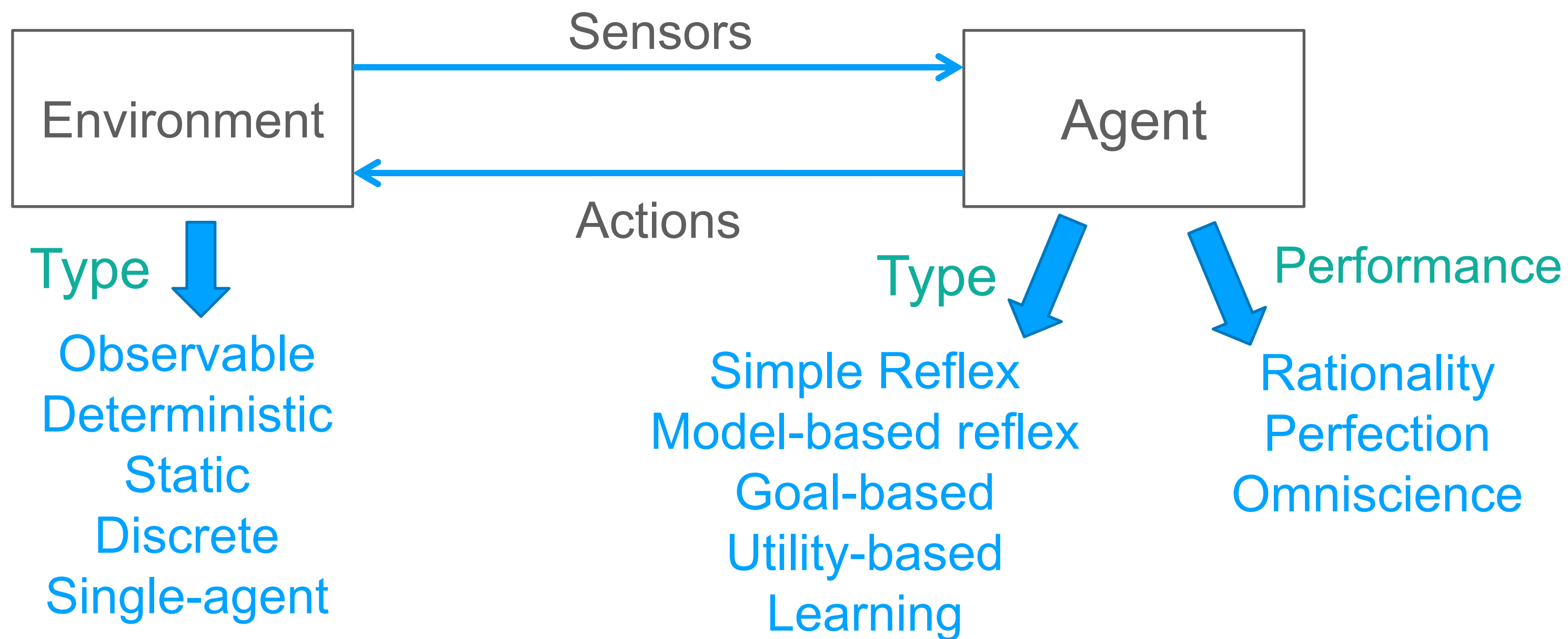
3+ days late → **0 points**

Homework solutions must be **handwritten**. You may either write directly on a tablet or write on paper and scan your work. In both cases, the final submission must be compiled into a single **PDF** file.

What is AI?

Rational Behavior

Doing the right thing (Expected to maximize goal achievement)



Environment properties

Fully observable / Partially observable

- **Fully observable:** the sensors can detect all aspects that are relevant to the choice of action
- **Partially observable:** The agent cannot observe the complete state of the environment.

Single-agent / Multi-agent

Single-agent: Only one agent operates in the environment.

Multi-agent: Multiple agents operate in the environment.

Environment properties

Deterministic / Stochastic

Deterministic: The next state is completely determined by the current state and the agent's action.

Stochastic: The next state involves randomness or uncertainty.

Sequential / Episodic

Sequential: Current actions affect future states and decisions.

Episodic: Each decision is independent of previous ones.

Environment properties

Static / Semi-dynamic / Dynamic

Static: The environment remains unchanged while the agent is deciding what to do.

Semi-dynamic: The environment itself does not change, but the agent's performance is affected by time.

Dynamic: The environment may change while the agent is thinking or taking actions.

Discrete / Continuous

Discrete: States, actions, or time are represented by discrete values.

Continuous: States or actions are represented by continuous values.

Exercise 1.18

Examine the AI literature to discover whether the following tasks can currently be solved by computers:

1. Playing a decent game of table tennis (Ping-Pong).
2. Driving in the center of Cairo, Egypt.
3. Driving in Victorville, California.
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8. Writing an intentionally funny story.
9. Giving competent legal advice in a specialized area of law.
10. Translating spoken English into spoken Swedish in real time. - Performing a complex surgical operation.

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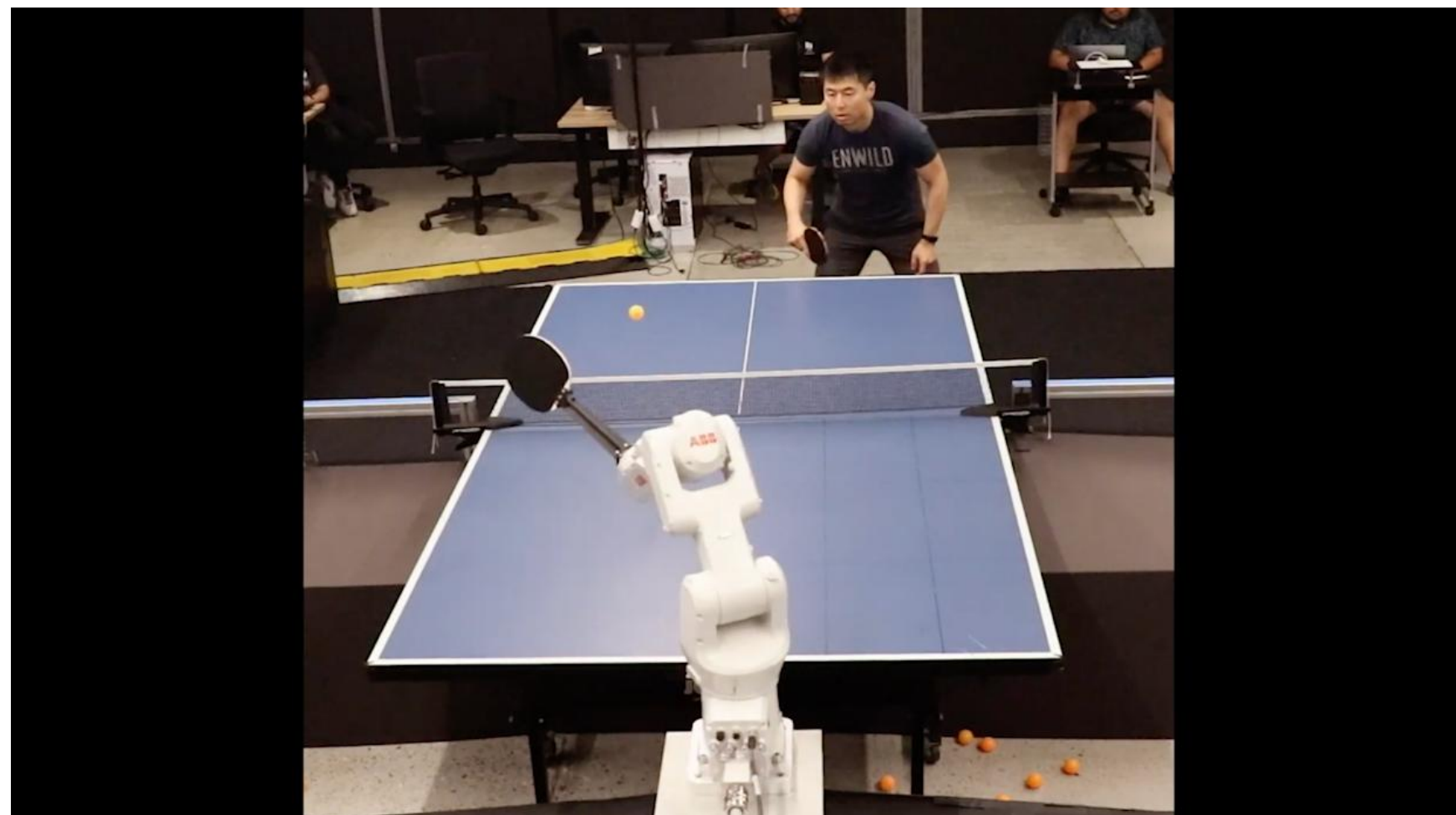
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Yes.

- bounded court, simple ball physics, short-horizon control.
- for example, a reasonable level of proficiency was achieved by Andersson's robot (Andersson, 1988), Google DeepMind Robot.



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No.

- no clear lanes, mixed vehicles, sudden crossings, blocked views



Image Source: <https://www.voanews.com/a/3762122.html>

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Yes*.

- marked lanes/signs, light traffic, good weather
- drivers follow rules



Image Source: <https://www.victorvilleca.gov/>

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Exercise 1.18

No.

- crowds + unseen items + squishy produce
- hard to combine all skills



Image Source: <https://www.pymnts.com/artificial-intelligence-2/2025/ai-powered-robots-advance-in-general-tasks-in-a-crowded-market/>

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Yes.

- forms/buttons, stable site patterns
- no physical manipulation

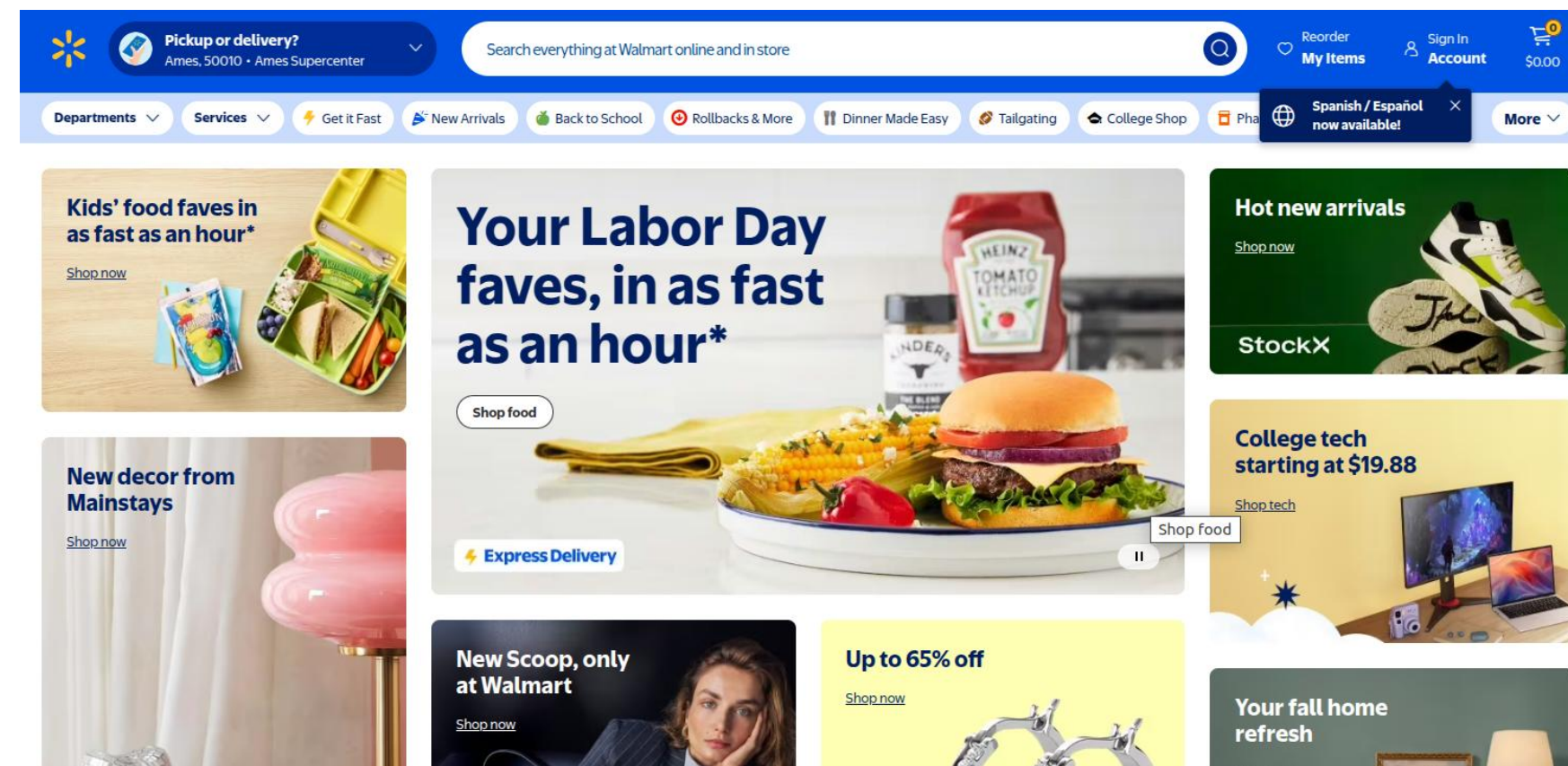


Image Source: <https://www.walmart.com/>

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Yes.

- fixed rules
- search + heuristics → strong play (programs already good)



Image Source: <https://www.funbridge.com/>

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Yes.

- within certain logics
- automated tools can prove some theorems

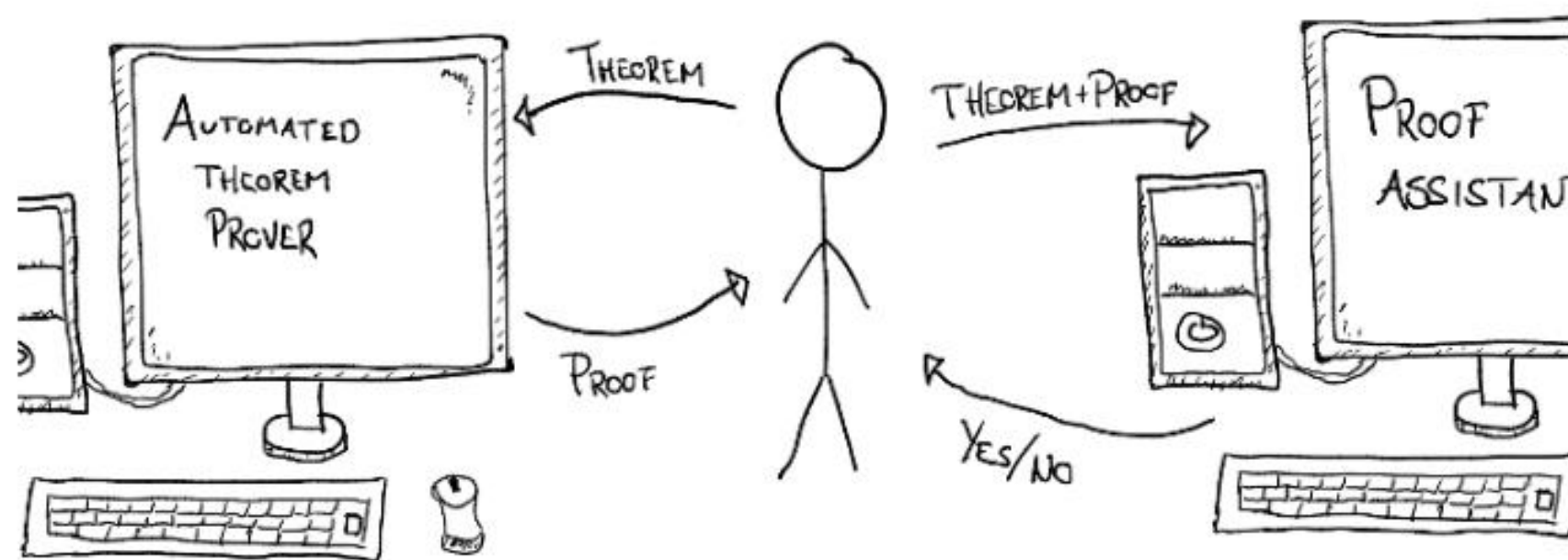


Image Source: <https://ecmiindmath.org/2023/02/16/automated-theorem-provers-and-their-role-in-cryptography/>

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No.

- needs people-sense, timing, shared culture
- hard to get right on purpose

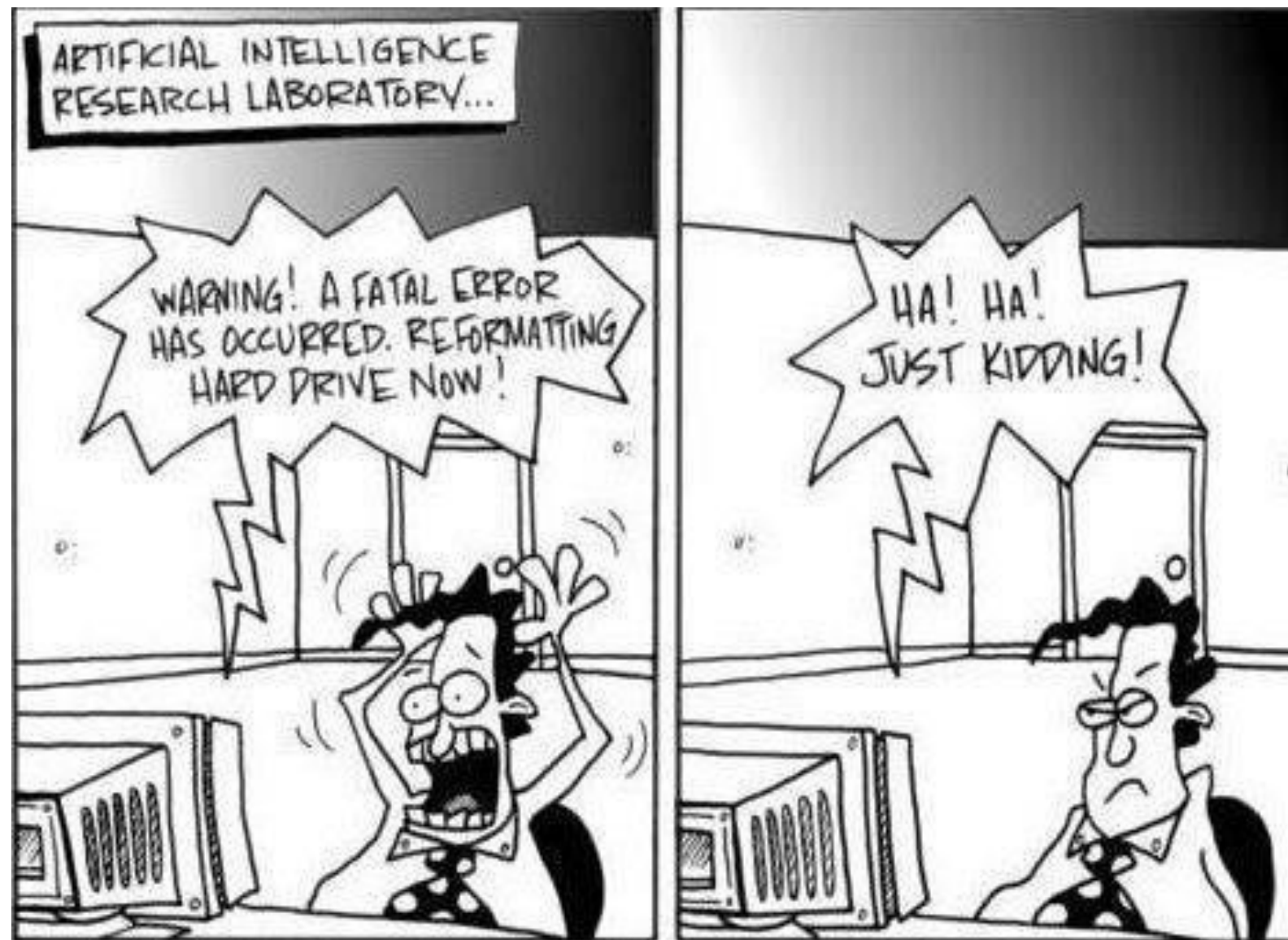


Image Source: <https://davidol.medium.com/will-ai-ever-be-able-to-make-a-joke-808a656b53a6>

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- narrow, rule-coded areas work
- broad cases need common knowledge & facts

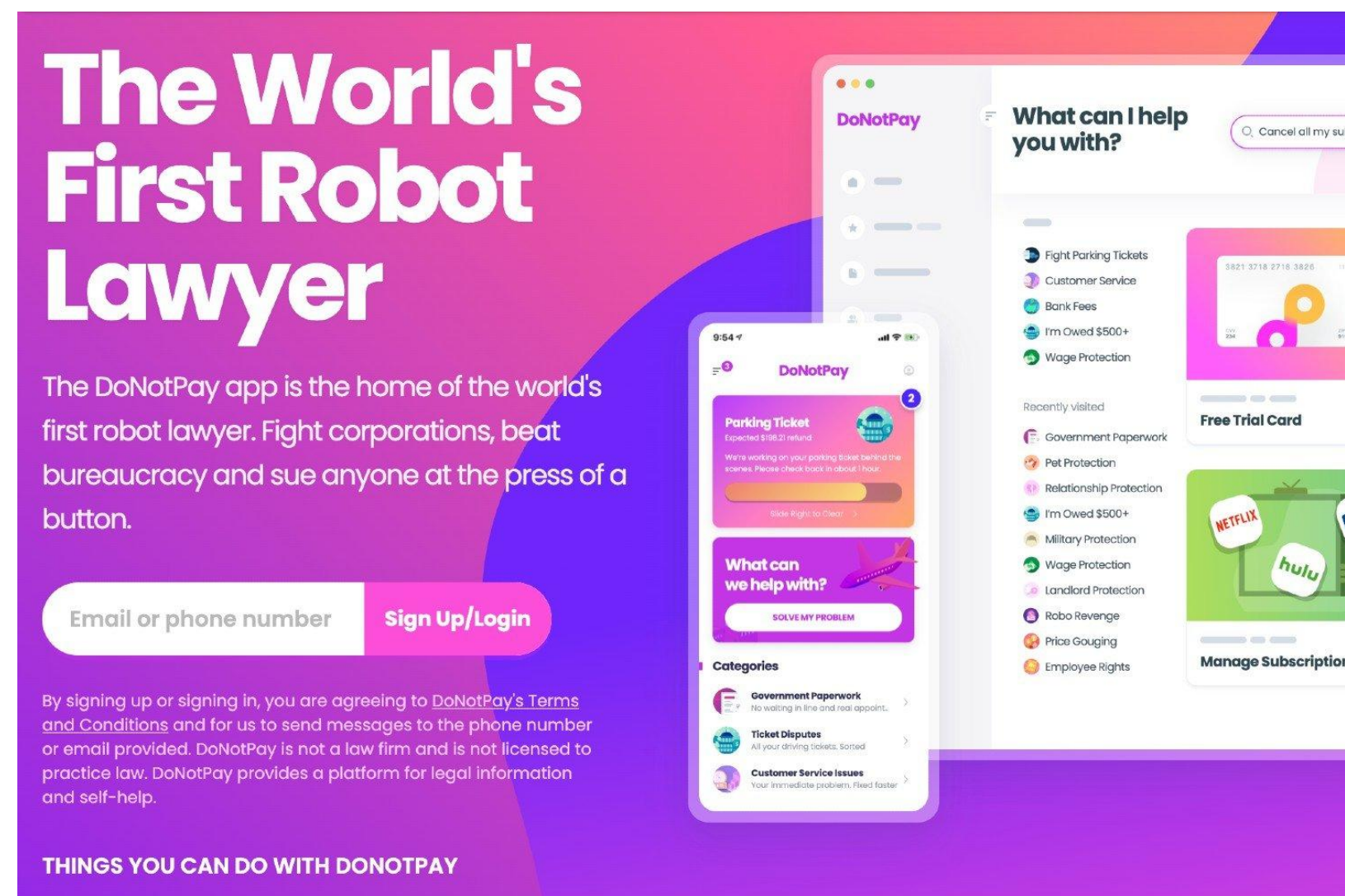


Image Source: <https://donotpay.com/>

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Yes*.

- strong for common language pairs/plain topics
- weaker on idioms & jokes

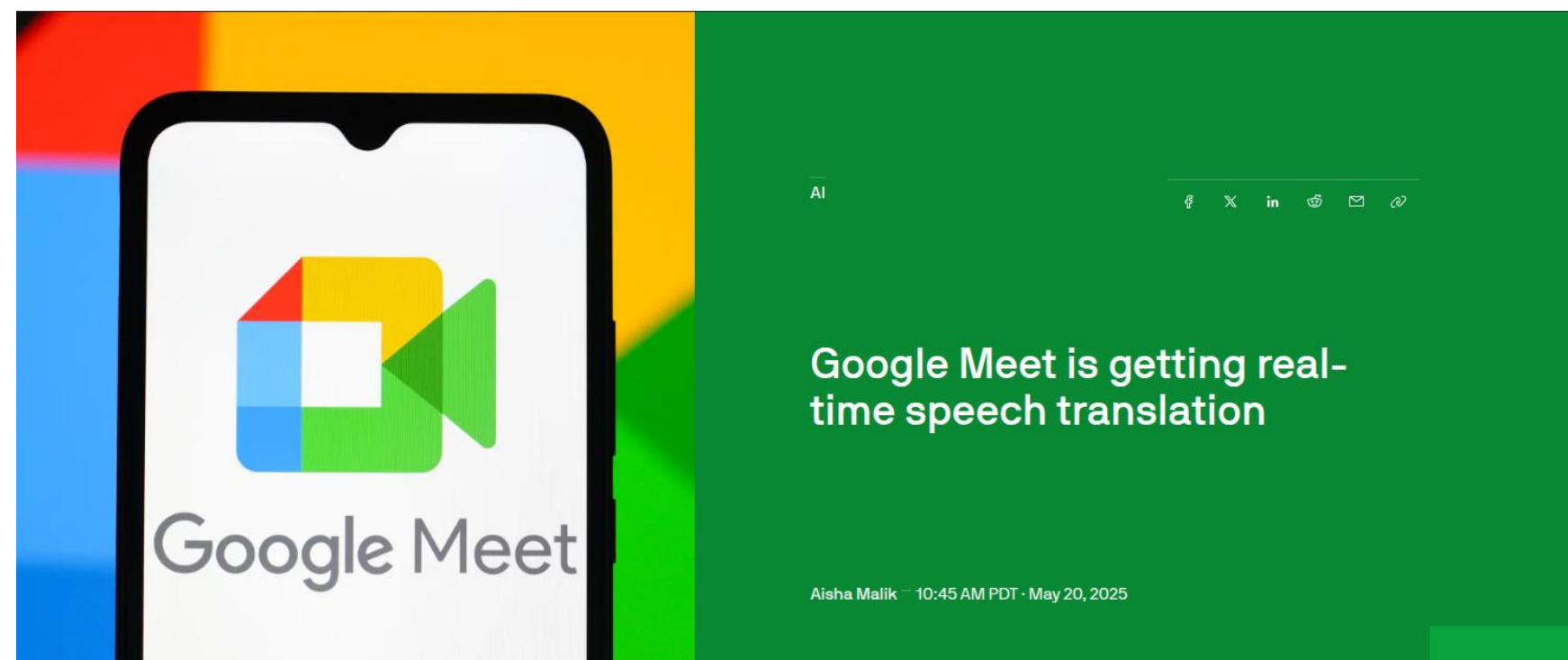


Image Source: <https://techcrunch.com/2025/05/20/google-meet-is-getting-real-time-speech-translation/>

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“That’s all Folks!”

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